Venkata Sandeep Kumar Reddy

DATA ENGINEER | CLOUD DATA | STREAMING | ANALYTICS ENGINEERING

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PROFESSIONAL SUMMARY

Data Engineer with approximately three years of professional experience building and supporting SQL/Python data workflows. Complementary hands-on engineering projects cover real-time streaming, change data capture (CDC), lakehouse workflows, and analytics engineering. Core strengths include data ingestion and transformation, validation and reconciliation, distributed processing, dimensional modeling, and preparing reliable data for analytics. Project results below include measured data volumes and recorded test/benchmark outcomes.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark

Data pipeline engineering: ETL/ELT, CDC, incremental processing, deduplication, idempotency, replay, checkpoints, data quality, reconciliation

Streaming and compute: Apache Kafka, Apache Spark, Spark Streaming, batch and distributed processing

Storage and modeling: PostgreSQL, Parquet, Apache Iceberg, lakehouse/warehouse concepts, dimensional modeling, slowly changing dimensions (SCD), data marts

Cloud and platform tooling: AWS S3, Glue Data Catalog, Athena, IAM, KMS, CloudWatch, Terraform, Airflow, Docker, Kubernetes, GitHub Actions, CI/CD

APIs, analytics and validation: FastAPI, React, Streamlit, scikit-learn, Isolation Forest, automated testing, observability, performance benchmarking

PROFESSIONAL EXPERIENCE

Data Engineer | Mar Symantech Private Limited

September 2023 – Present

- Build, maintain, and automate ETL/ELT workflows with Python and SQL to move structured and semi-structured data into downstream databases and analytical environments.
- Apply data-cleaning, schema-validation, and deduplication checks to improve data consistency and reduce the likelihood of invalid records reaching reporting and analytics consumers.
- Schedule and monitor batch jobs with Airflow and cron; investigate failures, trace processing problems, and support recovery of data workflows.
- Review SQL queries and execution plans; use indexing and transformation improvements to address query and processing bottlenecks.

- Prepare analytics-ready datasets and data marts, and document data definitions and lineage to support maintainable reporting.

To make this experience section more compelling, add only verified job-specific scale or impact (for example, pipeline count, data volume, runtime improvement, or stakeholder use). No such figures have been assumed here.

SELECTED ENGINEERING PROJECTS

Enterprise AI-Powered Real-Time EV Fleet Data Platform

Python · Apache Kafka · Apache Spark · PostgreSQL · FastAPI · Docker · Kubernetes

Project focus: A real-time telemetry data platform to ingest, process, validate, and expose electric-vehicle fleet data for analytical and anomaly-detection workflows.

- Processed 98,061,214 telemetry records, comprising 98,060,402 valid canonical records and 812 rejected records.
- Produced 160,774 feature windows and 8,251 AI samples for downstream analysis.
- Evaluated Isolation Forest anomaly detection with a recorded F1 score of 0.8619.
- API benchmark recorded 20/20 successful requests and 77.278 ms p95 latency.
- Project implementation covered streaming/compute, persistence, API access, containerization, and deployment-oriented engineering practices.

Cloud-Native Lakehouse, CDC & AI-Ready Data Platform

PostgreSQL pgoutput CDC · Raw/Bronze/Silver/Gold layers · Incremental processing

Project focus: A CDC-based data workflow that captures database changes and organizes data into layered datasets for downstream analytical use.

- Implemented and tested INSERT, UPDATE, and DELETE change events, including LSN progression and event ordering.
- Tested deduplication, idempotent processing, checkpoints, and replay-related behavior for reliable incremental handling.
- Recorded 15/15 automated tests, 63/63 data-quality checks, and 60/60 reconciliation checks.
- Validated 9 Raw, 9 Bronze, and 6 Gold datasets/products.

This description states tested project behavior; it does not imply a fully deployed production AWS environment.

RetailIQ — Modern Data Warehouse & Analytics Engineering Platform

PostgreSQL ELT · Dimensional modeling · Incremental models · SCD · Data marts · Streamlit

Project focus: Transforming source data into modeled analytical datasets and dashboard-ready products.

- Built incremental analytical models and BI-ready outputs using fact/dimension structures, slowly changing dimensions, data marts, and semantic views.
- Validated the work with 34/34 data-quality checks and 9/9 integration tests.
- Delivered a six-page Streamlit dashboard for presenting analytical outputs.

EDUCATION

B.Tech, Data Science — Presidency University (2019–2023)

PGP, Data Science and Machine Learning — Intellipaat (2024)

TRAINING & ENGINEERING PRACTICES

Linux training/certification — Intellipaat (December 2024). Data Science course — Intellipaat (January 2023).

Project engineering practices include automated testing, data-quality gates, reconciliation, observability, performance benchmarking, containerized services, and CI/CD.

Review before publishing: confirm employer’s exact legal name and dates, training names/dates, and that all skills and project statements are accurate and interview-defensible. Training entries were consolidated to avoid listing Linux training twice.